EC2 Basics

What is Amazon EC2?

- Ec2 is one of the most popular services in AWS/Cloud
- EC2 → Elastic Compute Cloud → Infrastructure as a Service
- What EC2 consists of
  - Renting Virtual Machines (EC2) → Instances
  - Storing Data on Virtual Drives (EBS)
  - Distributing Load across machines (ELB)
  - Scaling the services using an Auto-Scaling Group (ASG)

  → **Knowing EC2 is Fundamental to understand how the cloud works**

EC2 Sizing & Configuration Options

- **Operating System (OS):** Linux, Windows, & Mac OS
- How much Compute Power & Cores (CPU)
- How much Random-Access memory (RAM)
- How much Storage Space:
  - Network Attached (EBS & EFS)
  - Hardware (EC2 Instance Store)
- Network Card: Speed of the card, Public IP Address
- Firewall Rules: **Security Group**
- Bootstrap Script (Configure at First Launch): Ec2 User data

EC2 User Data

- It is possible to bootstrap our instances using an **EC2 User Data** Script
- Bootstrapping means launching commands when a Machine starts
- That script is only run-once at the instance First start
- EC2 user data is used to automate boot tasks such as:
  - Installing Updates
  - Installing Softwares
  - Download Common files from the internet
  - Anything you can think of

  → **EC2 User Data Script runs with the root user**

EC2 Instance types: Example

Instance	vCPU	Mem (GiB)	Storage	Network performance	EBS Bandwidth (Mbps)
t2.micro	1	1	EBS Only	Low to Moderate	
T2.xlarge	4	16	EBS Only	Moderate	
C5d.4xlarge	16	32	1 x 400 NVMe SSD	Upto 10 Gbps	4,750
R5.16xlarge	64	512	EBS Only	20 Gbps	13,600
M5.8xlarge	32	128	EBS Only	10 Gbps	6,800

Choose the instance that fits best for your application

EC2 Instance Types

Overview

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The naming convention of EC2:

Example: m5.2xlarge

m = Instance Class

5 = generation of instance

2xlarge = Size within the instance class

Instance Types – general purpose

- Great for a diversity of work loads such as web servers or code repositories
- Balance between
  - Compute
  - Memory
  - Networking
- We will use t2.micro which is General purpose

Instance Types – Compute optimised

- Great for compute intensive tasks that require high performance processors
  - Batch Processing workloads
  - Media Transcoding
  - High Performance web Servers
  - High Performance Computing
  - Scientific Modelling & machine Learning
  - Dedicated Gaming Servers

Instance Types – Memory optimised

- Fast performance for workloads that process large data sets in memory
- Use cases:
  - High performance, Relational/non relational databases
  - Distributed web scale cache stores
  - In-memory databases optimized for BI (Business Intelligence)
  - Applications performing real-time processing of big unstructured data

Instance Types – Storage optimised

- Great for storage-intensive tasks that require high, sequential read and write access to large data sets on local storage
- Use cases:
  - High Frequency online transaction processing (OLTP) systems
  - Relational & NoSQL databases
  - Cache for in-memory databases (for example, Redis)
  - Data Warehousing applications
  - Distributed file systems